5/4/24 NLPP

Non Linear Programming Problem

* We shall consider 3 types of NLPP :

1. Quadratic PP with no constraints
2. Quadratic PP with linear equality constraints
3. Quadratic PP with linear inequality constraints

1. Quadratic PP with no constraints

→ Object fn is of the form :

$$Z = a_{11}x_1^2 + a_{22}x_2^2 + \dots + a_{nn}x_n^2 + a_{12}x_1x_2 + a_{13}x_1x_3 + \dots + a_{1n}x_1x_n + \dots$$

Put $\frac{\partial f}{\partial x_i} = 0 \quad 1 \le i \le n$

& find $X_0 \; (x_1, x_2 \dots x_n)$

Define a Hessian matrix

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

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PAGE No.
DATE / /

→ If 1) If all the principle minor determinants of Hessian matrix are +ve (at X_0) then X_0 is a ~~Maxima~~ Minima

2) If the principle minor det. $D_1, D_3 \ldots$ are −ve and $D_2, D_4 \ldots$ are +ve then X_0 is a Maxima

3) In general, if S_n matrix is indefinite (random) at X_0 and X_0 is saddel pt.

eg Optimise $Z = x_1^2 + x_2^2 + x_3^2 - 6x_1 - 8x_2 - 10x_3$

→ Let $Z = f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 - 6x_1 - 8x_2 - 10x_3$

$$\frac{\partial f}{\partial x_1} = 0 \Rightarrow 2x_1 - 6 = 0 \qquad x_1 = 3$$

$$\frac{\partial f}{\partial x_2} = 0 \Rightarrow 2x_2 - 8 = 0 \qquad x_2 = 4$$

$$\frac{\partial f}{\partial x_3} = 0 \Rightarrow 2x_3 - 10 = 0 \qquad x_3 = 5$$

$\therefore$ $x_0 (3, 4, 5)$ is a stationary pt

at X_0,

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} & \frac{\partial^2 f}{\partial x_3 \partial x_1} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

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$D_1 = |[2]| = 2$

$$D_2 = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4$$

$$D_3 = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 8$$

Since D_1, D_2, D_3 all are +ve, x_0 is a pt of minima.

$$\therefore Z_{min} = 3^2 + 4^2 + 5^2 - 6(3) - 8(4) - 10(5)$$
$$= 9 + 16 + 25 - 18 - 32 - 50$$
$$= -50$$

7/4/24

Q. Obtain relative maxima or minima of the fn

$$Z = x_1 + 2x_3 + x_2 x_3 - x_1^2 - x_2^2 - x_3^2$$

→ Let $f(x_1, x_2, x_3) = x_1 + 2x_3 + x_2x_3 - x_1^2 - x_2^2 - x_3^2$

$\frac{\partial f}{\partial x_1} = 0 \Rightarrow x_1 = 1/2$ $[1 - 2x_1 = 0]$

$\frac{\partial f}{\partial x_2} = 0 \Rightarrow x_2 = 2/3$ $[x_3 - 2x_2 = 0]$

$\frac{\partial f}{\partial x_3} = 0 \Rightarrow x_3 = 4/3$ $[2 - 2x_3 + x_2 = 0]$

$\therefore X_0 \left(1/2, 2/3, 4/3\right)$ is a stationary pt

consider $H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix}$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

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consider $D_1 = |[-2]| = -2$

$$D_2 = \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix} = 4$$

$$D_3 = \begin{vmatrix} -2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & -2 \end{vmatrix} = -6$$

Since D_1 & D_3 are -ve & D_2 is +ve,
X_0 is a pt of maxima

$$Z_{max} = \frac{1}{2} + 2 \times \frac{4}{3} + \frac{2}{3} \times \frac{4}{3} - \frac{1}{4} - \frac{4}{9} - \frac{16}{9}$$

$$= \frac{1}{2} + \frac{8}{3} + \frac{8}{9} - \frac{1}{4} - \frac{4}{9} - \frac{16}{9}$$

$$= \frac{19}{12}$$

2. NLPP with equality constraints

(i) With 1 equality constraint :

Consider

$Z = f[x_1, x_2 \dots x_n]$

sub to $g(x_1, x_2 \dots x_n) = b$

i.e $g(x_1, x_2 \dots x_n) - b = h(x_1, x_2 \dots x_n) = 0$

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Construct a new function called Lagrangian function using Lagrangian multipliers

$$L(x_1, x_2 \ldots x_n, \lambda) = f(x_1, x_2 \ldots x_n) - \lambda h(x_1, x_2 \ldots x_n) \quad \text{—(1)}$$

The necessary cond^n for maxima/minima subject to the constraint $h(x_1, x_2 \ldots x_n) = 0$ are,

$$\left.\begin{aligned} \frac{\partial L}{\partial x_1} &= 0 \\ \frac{\partial L}{\partial x_2} &= 0 \\ &\vdots \\ \frac{\partial L}{\partial x_n} &= 0 \\ \frac{\partial L}{\partial \lambda} &= 0 \end{aligned}\right\} \text{—(2)}$$

from (1), we get

$$\frac{\partial L}{\partial x_1} = \frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1}$$

$$\frac{\partial L}{\partial \lambda} = -h$$

Using (2) we get following $n+1$ cond^{n}

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$$\cdots$$

$$h(x_1, x_2 \ldots x_n) = 0$$

Solving these $n+1$ cond^{n}s, we find x_1, x_2, x_n and λ and find a pt of maxima / minima

Now find the value of the following det of order $n+1$ at x_0.

$$\Delta_{n+1} = \begin{vmatrix} 0 & \frac{\partial h}{\partial x_1} & \partial h / \partial x_2 \\ \frac{\partial h}{\partial x_1} & \frac{\partial^2 f}{\partial x_1^2} - \lambda \frac{\partial^2 h}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} - \lambda \frac{\partial^2 h}{\partial x_1 \partial x_2} \\ \frac{\partial h}{\partial x_2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} - \lambda \frac{\partial^2 h}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} - \lambda \frac{\partial^2 h}{\partial x_2^2} \\ \vdots & \vdots & \vdots \\ \frac{\partial h}{\partial x_n} & \frac{\partial^2 f}{\partial x_n \partial x_1} - \lambda \frac{\partial^2 h}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} - \lambda \frac{\partial^2 h}{\partial x_n \partial x_2} \end{vmatrix}$$

.... n col

$\boxed{n \geq 1}$

Teacher's Signature:

PAGE No.
DATE / /

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→ If the signs of the principle minors $\Delta_3, \Delta_4, \Delta_5 \dots$ are alternatively +ve and −ve, the pt X_0 is a pt of maxima.

→ If all the principle minors $\Delta_3, \Delta_4, \Delta_5 \dots \Delta_{n+1}$ are −ve, then X_0 is a pt of minima.

Q Use a method of Lagrange multiplier to solve the following NLPP.

$$Z = 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 - 100$$

subj to

$$x_1 + x_2 + x_3 = 20$$
$$x_1, x_2, x_3 \geq 0$$

→ $L(x_1 x_2 x_3 \lambda) = f(x_1 x_2 x_3) - \lambda h(x_1 x_2 x_3)$

$$= 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 - 100 - \lambda(x_1 + x_2 + x_3 - 20)$$

$$\frac{\partial L}{\partial x_1} = 0 \Rightarrow 4x_1 + 10 - \lambda = 0 \quad -(1)$$

$$\frac{\partial L}{\partial x_2} = 0 \Rightarrow 2x_2 + 8 - \lambda = 0 \quad -(2)$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow 6x_3 + 6 - \lambda = 0 \quad -(3)$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow x_1 + x_2 + x_3 = 20 \quad -(4)$$

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Multiply (1) by 3, (2) by 6, (3) by 2 & use eqn (4)

$$12(\underbrace{x_1 + x_2 + x_3}_{20}) + 90 - 11\lambda = 0$$

$$240 + 90 = 11\lambda$$

$$\frac{330}{11} = \lambda$$

$$\boxed{\lambda = 30}$$

$\therefore$ from (1), (2), (3) [subs $\lambda = 30$]

$x_1 = 5$

$x_2 = 11$

$x_3 = 4$

$(Z_{min} = 281)$

$\therefore$ $X_0 (5, 11, 4)$ is a stationary pt

$$\Delta_4 = \begin{vmatrix} 0 & \partial h/\partial x_1 & \partial h/\partial x_2 & \partial h/\partial x_3 \\ \frac{\partial h}{\partial x_1} & & & \frac{\partial^2 f}{\partial x_1 \partial x} - \lambda \frac{\partial^2 h}{\partial x_1 \partial x_3} \\ \frac{\partial h}{\partial x_2} & & & \frac{\partial^2 f}{\partial x_2 \partial x_3} - \lambda \frac{\partial^2 h}{\partial x_2 \partial x_3} \\ \frac{\partial h}{\partial x_3} & & & \frac{\partial^2 f}{\partial x_3^2} - \lambda \frac{\partial^2 h}{\partial x_3^2} \\ \cancel{\frac{\partial h}{\partial x}} & & & \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 4 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 6 \end{vmatrix} \Rightarrow \begin{matrix} C_2 - C_4 \\ C_3 - C_4 \end{matrix}$$

$$\therefore \Delta_4 = \begin{vmatrix} 0 & 0 & 0 & 1 \\ 1 & 4 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & -6 & -6 & 6 \end{vmatrix} = -1 \begin{vmatrix} 1 & 4 & 0 \\ 1 & 0 & 2 \\ 1 & -6 & -6 \end{vmatrix}$$

$$= -44.$$

$$= C_2 - 4C_1$$

$$\Delta_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 4 & 0 \\ 1 & 0 & 2 \end{vmatrix} = -6$$

Since Δ_3 & Δ_4 are both −ve, X_0 is a pt of minima

$\therefore Z_{min} = 281$

2	4	2	6
2	2	1	3
3	1	1	3

2	12	8	6
2	6	4	3
2	3	2	3
3	3	1	3
	1	1	1

PAGE No.
DATE / /

NLPP with one inequality constraint

Kuhn Tucker method

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$$h(x_1, x_2) \le 0$$

$$\lambda h(x_1, x_2) = 0$$

$$\lambda \geq 0$$

q

$p = 0.55$ (

$n = 10$

$^{n}C_x p^x q^{n-x}$

A : iphone owners

B : men

X : iphone owners selected are men

$P(X = 7) = {}^{10}C_7 (0.55)^7$

$P(A/B) = 55\% = 0.55$

$P($

$p = 0.55$ (iphone & men)

$q = 1 - p = 0.45$

$^{n}C_x p^x q^{n-x}$

Alt soln ⇒ when value of non basic variable in z row is 0.

NOTES

$\Sigma x = 130$ $\quad \bar{x} = \frac{130}{10} = 13$

$\Sigma y = 220$

Σx^2 $\quad \bar{y} = \frac{220}{10} = 22$

$$b_{yx} = \frac{\Sigma xy - N\bar{x}\bar{y}}{\Sigma x^2 - N\bar{x}^2}$$

$$= \frac{3467 - 10(13)(22)}{2288 - 10(169)}$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$=$$

$$\frac{7 \times 5 \times 4}{3 \times 2}$$

Sample size	σ ⇒ known	σ ⇒ unknown
$n \geq 30$	Z-test	Z-test
$n < 30$	Z-test	t-test

if ratio same $\Rightarrow r = R$

t $\Rightarrow$ small sampling & when std deviation is not known

$\sigma \Rightarrow$ known $\Rightarrow$ z test

NOTES

$\frac{\partial^2 f}{\partial x^2} = r$

$\frac{\partial^2 f}{\partial x \partial y} = s$

$\frac{\partial^2 f}{\partial y^2} = t$

$\left.\begin{array}{l} rt - s^2 > 0 \\ r > 0 \end{array}\right\}$ Maxima / Minima

$\left.\begin{array}{l} rt - s^2 > 0 \\ r < 0 \end{array}\right\}$ max

Karl Pearson's Coeff of correlation $\Rightarrow$ linear relationship b/w continuous variables.

$$\left(r = \frac{cov(x, y)}{\sigma_x \times \sigma_y} \right)$$

Spearman's rank correlation coeff $\Rightarrow$ for ordinal, non linear non normally distributed data.

$$\left(R = 1 - \boxed{\frac{6 \sum d_i^2}{n^3 - n}} \right)$$

sample size	σ known	σ unknown
$n \geq 30$	z test	z test
$n < 30$	z test	t test

optimal basic feasible soln

↳ artificial variable leave the process & optimality condn is satisfied by the basic variable

Degenerate

↳ atleast one of the artificial variable remains in the basis with zero value & optimality condn is satisfied.

χ^2 dist $\Rightarrow$ non parametric